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The 2022-2023 Amoxicillin Shortage and Acute Otitis Media Treatment

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INTRODUCTION

High-dose amoxicillin is the first-line therapy for acute otitis media (AOM) — the most common indication for antibiotics in children.[1] On October 28th, 2022, the U.S. Food and Drug Administration (FDA) announced a national shortage of amoxicillin liquid suspension.[2] Outbreaks of seasonal respiratory viruses starting in autumn 2023 have increased demand; however, most manufacturers did not disclose a reason for the shortage or resolution timeframe.[3] More broadly, the extent to which drug shortages affect prescribing patterns has not been well-characterized.[4] The American Academy of Pediatrics recommended various alternatives to treat AOM, including splitting tablet or capsule formulations, use of traditionally second-line agents, and watchful waiting.[5]

To characterize the causal impact of an unexpected shock to medication availability, we performed a quasi-experimental analysis of antibiotic prescribing for uncomplicated AOM. We hypothesized that amoxicillin prescribing would immediately decrease alongside greater use of broader-spectrum antibiotics, and a gradual return to baseline prescribing patterns.

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Ryan Brewster conceptualized and designed the study, collected data, carried out the initial analyses, and drafted and revised the manuscript
Rohan Khazanchi conceptualized and designed the study, analyzed and interpreted the data, and drafted and revised the manuscript
Alex Butler conceptualized and designed the study, collected data, and drafted and revised the manuscript
Daniel O'Meara conceptualized and designed the study and drafted and revised the manuscript
Devika Bagchi drafted and revised the manuscript
Kenneth Michelson coordinated and supervised the data collection, analysis, and interpretation, and critically reviewed the manuscript.
All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

METHODS

We retrospectively analyzed encounters for children <18 years old diagnosed with AOM between May 15th, 2022 and April 14th, 2023 at an urban freestanding children's hospital system and affiliate community health center. Our exclusion criteria were a previous episode of AOM within 30 days, concomitant respiratory infections that might alter treatment (sinusitis, pharyngitis, pneumonia, and conjunctivitis), and an amoxicillin or penicillin allergy (ICD-10 codes in eAppendix 1).

We described patient demographic characteristics by pre/post-shortage period. We used a sharp regression discontinuity design to assess whether prescribing changed immediately after the FDA's declaration (October 28th, 2022).[6] This approach used logistic regressions to evaluate the causal impact of the shortage on the odds of being prescribed amoxicillin, amoxicillin-clavulanate, cefdinir, other antibiotics, or no antibiotic. Analysis details and sensitivity checks are described in eAppendices 2/3. Of note, racial and ethnic categories were included as covariates to evaluate differential impacts of the amoxicillin shortage resulting from known inequities in access to prescription medications. The study was exempted by the Boston Children's Hospital Institutional Review Board.

RESULTS

We identified 3076 encounters for AOM, with 1677 (54.5%) occurring after the amoxicillin shortage (Table 1). The median patient age was 3.1 years (IQR 1.6-5.4), and most patients were male (n=1695, 55.1%), Hispanic (n=1252, 40.7%), and evaluated in the emergency department (n=1900, 61.8%). The most prescribed antibiotics were amoxicillin (n=1662, 54.0%), amoxicillin-clavulanate (n=671, 21.8%), and cefdinir (n=263, 8.6%).

The odds of being prescribed amoxicillin decreased by 91% after the FDA shortage declaration (OR 0.09 [95%CI 0.07-0.12]) (Figure 1A). In contrast, the odds of being prescribed amoxicillin-clavulanate or cefdinir increased 7-fold and 9-fold (7.90 [5.59-11.31] and 9.25 [5.27-17.17]), respectively (Figure 1B/1C). Management without antibiotics (0.82 [0.55-1.23]) and prescriptions for other antibiotics (1.73 [0.96-3.19]) did not significantly change (Figure 1D/1E). By the end of the study period, antibiotic prescribing approached pre-shortage levels. Trends in AOM management were similar across sociodemographic characteristics (eAppendix 4)

DISCUSSION

The 2022-2023 amoxicillin shortage had significant and immediate impacts on antibiotic choice for AOM. As amoxicillin prescribing declined, the use of alternative agents like amoxicillin-clavulanate and cefdinir sharply increased. Rates of watchful waiting remained unchanged and lower than anticipated, which may suggest a propensity to treat higher acuity patients in the emergency department.

Antimicrobial and other drug shortages have become increasingly common consequences of limited investment in low-profit, generic medications and regulatory challenges, among other factors. [7,8] Our study illustrates one direct and previously underexplored impact: how dramatically clinical practice responds to market disruptions. While amoxicillin-clavulanate and cefdinir have comparable efficacy to amoxicillin for AOM, they are less cost-effective, carry worse side effect profiles, and exhibit broader spectrum activity, raising concerns for economic impacts and downstream antimicrobial resistance.[9] Antibiotic shortages may also contribute to medication errors, adverse events, and treatment delays. [10]

Study limitations include the retrospective, single-center design; local constraints may have varied by drug supplier, region, or rurality. The lack of differences across sociodemographic characteristics likely reflects the small catchment area and uniform pharmacy availability. That said, our rigorous quasi-experimental methods and sensitivity checks improve the validity of our findings. Another limitation is that we did not examine prescribing trends for other respiratory infections, but anticipate that similar shifts occurred.

Drug shortages have an immediate, sweeping effect on prescribing patterns and should be monitored and intervened upon by regulatory agencies, policymakers, and health systems alike. The FDA should consider increasing oversight of essential medications, requiring disclosure of supply issues, and incentivizing antibiotic production to mitigate their low profitability.[4]

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Conflict of Interest Disclosures:

Dr. Khazanchi reported previously serving as a health equity consultant to the New York City Department of Hygiene and Mental Health’s Office of the Chief Medical Officer; currently serving as a strategic advisory board member for the Rise to Health Coalition; and receiving grant funding from Boston Children’s Hospital, all outside the submitted work. Dr. Michelson reports receiving grant funding from the Agency for Healthcare Research and Quality. The other authors have no conflicts of interest to disclose.

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Abbreviations:

AOM	acute otitis media
FDA	U.S. Food and Drug Administration

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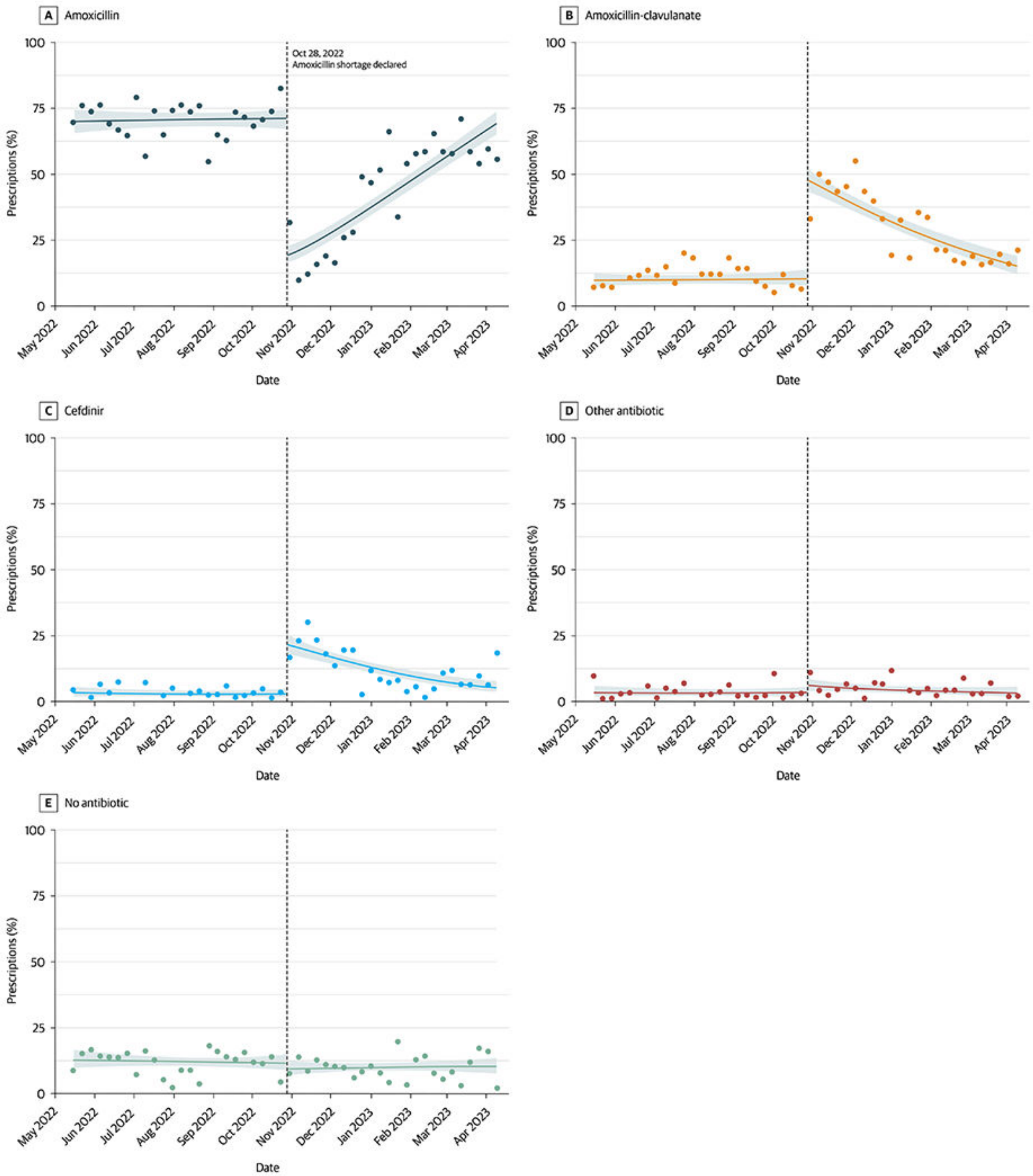

Figure 1.

Regression Discontinuity Plot of Antibiotic Prescriptions Before and After the FDA Amoxicillin Shortage Declaration (October 28, 2022).

Table 1.

Encounter Characteristics for Patients Diagnosed with Acute Otitis Media During the Pre-Shortage (May 15, 2022 - October 27, 2022) and Shortage (October 28, 2022 - April 14, 2023) Period.

	Pre-Shortage (n=1399)	Shortage (n=1677)	Overall (n=3076)
Prescription, n (%)			
Amoxicillin	991 (70.8%)	671 (40.0%)	1662 (54.0%)
Amoxicillin-clavulanate	140 (10.0%)	531 (31.7%)	671 (21.8%)
Cefdinir	43 (3.1%)	220 (13.1%)	263 (8.6%)
No antibiotic	171 (12.2%)	169 (10.1%)	340 (11.1%)
Other [a]	54 (3.9%)	86 (5.1%)	140 (4.6%)
Race/ethnicity, n (%) [b]			
Hispanic	596 (42.6%)	656 (39.1%)	1252 (40.7%)
Black, non-Hispanic	287 (20.5%)	296 (17.7%)	583 (19.0%)
Unknown	159 (11.4%)	220 (13.1%)	379 (12.3%)
Other [c]	162 (11.6%)	233 (13.9%)	395 (12.9%)
White, non-Hispanic	153 (10.9%)	218 (13.0%)	371 (12.1%)
Asian, non-Hispanic	42 (3.0%)	54 (3.2%)	96 (3.1%)
Gender, n (%)			
M	770 (55.0%)	925 (55.2%)	1695 (55.1%)
F	629 (45.0%)	752 (44.8%)	1381 (44.9%)
Age, median (IQR)	2.96 (1.59-5.23)	3.26 (1.55-5.48)	3.13 (1.56-5.38)
Encounter location, n (%)			
Emergency department	849 (60.7%)	1051 (62.7%)	1900 (61.8%)
Clinic or urgent care	546 (39.0%)	623 (37.1%)	1169 (38.0%)
Unknown	4 (0.3%)	3 (0.2%)	7 (0.2%)
Insurance, n (%)			
Public	1030 (73.6%)	1163 (69.4%)	2193 (71.3%)
Private	267 (19.1%)	358 (21.3%)	625 (20.3%)
Public-private	86 (6.1%)	82 (4.9%)	168 (5.5%)
Other	11 (0.8%)	61 (3.6%)	72 (2.3%)
Unknown	5 (0.4%)	13 (0.8%)	18 (0.6%)

[a] Includes ceftriaxone (n=95, 3.1%), azithromycin (n=30, 1%), and others (n=15, 0.5%)

[b] Race and ethnicity information was obtained from patient/family-reported data at the time of registration, as documented in the electronic medical record

[c] Includes American Indian/Alaska Native, Native Hawaiian/Pacific Islander, multiracial, and other race children